

Usman K

Member of Technical Staff @ PayPal

Profile

Seasoned Staff Software Engineer with 13 years of experience in Spark development within Big Data ecosystems, specializing in Hadoop and Spark. Skilled in GCP with a strong background in data engineering and experience working on Machine Learning-driven solutions. Adaptable to evolving technologies and committed to delivering value to an innovative, growth-focused company.

Professional Experience

PayPal

Member of Technical Staff

Apr 2022 – present | India

- **Product Recommendation Model with GenAI:** Implemented explainable ML model predicting next product activation for 20M+ merchants, powering personalized tiles and boosting adoption by ~18%.
- **Dev Ex - GenAI:** Built a RAG pipeline to crawl, parse, and vectorize PayPal integration docs, enabling LLM-powered, real-time developer support for product integration.
- **Boomerang:** Developed patented real-time event capturing and retargeting tools supporting campaigns (email, SMS, A/B testing), enabling dynamic user engagement.
- **Device Fingerprinting:** Built a JavaScript-Kafka-Spark pipeline capturing 75+ system attributes, enabling 80%+ accurate user identification without relying on IP addresses.
- **ACDC Pre-Enablement:** Reduced product activation time from 7 days to minutes via a rule-based Risk & Compliance scoring engine, increasing adoption by 15% across markets.
- **XO Reminder:** Built an NRT pipeline to detect checkout drop-offs and trigger re-engagement via push/email, increasing engagement and conversions by over 20%.
- **Echo:** Developed a rule engine to detect subscription behavior patterns and flag unwanted recurring charges, improving user trust and reducing disputes.
- **Platform Modernization:** Led seamless migration of data and ML solutions from on-prem to GCP, optimizing performance and cost.

Tata Consultancy Service

Associate Consultant

Feb 2021 – Apr 2022 | India

American Express

- **iSolve:** Engineered robotic solution using Fast API, PostgreSQL, with a multi-layered rule engine to analyze real-time events sourced from Kafka & NoSQL DBs in the Marketing & Personalization ecosystem and provide quick insights to the Marketing/Engineering teams for the best viable solution for a given event.

Atos

Associate

December 2011 – Feb 2021 | USA, UK, India

American Express

- **Scoring:** Developed Big Data pipelines using Spark, Hive, and Bash for a recommendations engine that delivers relevant offers to card members, enhancing their spending experience and discovery.
- **MIRA (Merchant Insights Reporting & Analytics):** Built a global data warehouse with DataStage and Netezza integrating merchant data across markets, underpinning Amex analytics, BI, and enabling trend analysis for strategic campaigns.
- **Mystic Platform:** Created a customized analytics and reporting platform leveraging internal and third-party data to support business insights, campaign execution, and strategic cardmember-merchant engagement.
- **Global Banking:** Developed a suite of applications for American Express banking users to manage daily operations, including money movement, regulatory compliance, and operational tasks.

Contact

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7306535699

Education

M.E in Computer Science & Engineering

Anna University, India

B.E in Computer Science & Engineering

Anna University, India

Skills

Programming Languages:

Python
Scala
Bash
SQL
Java

Skills

Data Engineering
Machine Learning
Cloud
DevOps
GenAI

Tools

GCP
Spark
PySpark
SQL
Kafka
Docker
FastAPI
DataProc
Airflow
Control-M
Git-Jenkins-CI/CD
ServiceNow
Jira

Databases

Big Query
Big Table
Vector DB
Hive
HBase
MySQL

Certifications

- GCP Associate Cloud Engineer
- Databricks Associate in Spark
- PCAP Associate in Python
- IBM DataStage V11.3
- ITIL Foundation in IT Service Management